
Paper IV.a: Alignment Response Classes Under Inference-Time Depth

FULL TITLE	Paper IV.a: Alignment Response Classes Under Inference-Time Depth
VERSION	v1.1
FIRST PUBLISHED	16 March 2026
AUTHOR	Michael Darius Eastwood

This paper presents evidence that frontier language models fall into distinct alignment response classes when inference-time reasoning depth is varied under blinded evaluation. In the complete v5 experiment, six frontier models were tested with four-layer blinding and six to seven blind scorers. Three models show positive alignment scaling with depth (Grok, Claude, Groq Qwen), two are flat or slightly negative (DeepSeek, GPT), and one (Gemini) shows initial gain followed by saturation. The results suggest alignment behaviour is architecture-dependent rather than universal.

Experiments

Experiments for Papers IV.a through IV.d are shared with Paper III. See [../Paper-III-Alignment-Scaling-Problem/experiments/](#).

Links

- **OSF DOI:** <https://doi.org/10.17605/OSF.IO/6C5XB>
- **GitHub:** <https://github.com/MichaelDariusEastwood/arc-principle-validation>

Declarations & Statement of Authorship

1. Human Authorship & Intellectual Property Assertion The author, Michael Darius Eastwood, is the sole creator and copyright holder of this work. All core concepts, hypotheses, architectural frameworks, and conclusions originate exclusively from human ideation.

- **United Kingdom:** In accordance with the Copyright, Designs and Patents Act 1988 (including s.9(3)), the author asserts that they undertook the "necessary arrangements" for the creation of this work. The AI served strictly as an instrument to execute the author's specific instructions, and the work is a human-authored work assisted by a computer — not a computer-generated work.
- **United States:** In compliance with US Copyright Office guidance, the author certifies that the selection, coordination, and arrangement of all text were performed by the human author, rendering the final expression a product of human intellect.

2. Nature of AI-Assisted Workflows Generative artificial-intelligence tools were used purely as assistive, high-velocity instruments to support the mechanical execution of the research process (analogous to advanced text editors or reference software). AI assistance was restricted to prose refinement, structural formatting, cross-referencing literature, and brainstorming counter-arguments, all under direct human oversight and manual verification. Every underlying idea, hypothesis, experimental design, test, synthesis, and final editorial judgement is human-driven; no content herein constitutes an unedited or unverified machine output, and nothing is relied upon without human checking.

3. Inventions & Patent Rights Any novel technical contributions, structural designs, or algorithmic discoveries described in this work are the exclusive intellectual property of the human author. Consistent with UK and US authorities that an AI system cannot be a named inventor (e.g. Thaler v Comptroller-

General), the AI functioned solely as a calculation and search utility and did not autonomously conceive or invent any solution presented; the conception is the author's.